

Lecture 11 – Completeness for exponential families

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11.1 Complete sufficient statistics in exponential families

Recall that T is a complete sufficient statistic if $\mathbb{E}_\theta[a(T)] = 0$ implies $a(T) = 0$ with probability 1 for all θ . The following theorem shows the natural statistic in exponential families is always complete.

Theorem 1 (Completeness in a one-parameter exponential family). Suppose $X_1, \dots, X_n$ are drawn i.i.d. from an exponential family, with density or mass function

$$f_\theta(x) = h(x) \exp\{b(\theta)R(x) - c(\theta)\}.$$

Then

$$T = \sum_{i=1}^n R(X_i)$$

is sufficient and complete.

We have already proved sufficiency (in fact, minimal sufficiency) of T in Lecture 5. We state the completeness part without proof.

Example 1 (Finishing the Poisson example). In Lecture 9 on Rao-Blackwellization, we considered improving the unbiased estimator $U = \frac{1}{n} \sum_{i=1}^n \mathbb{1}\{X_i = 0\}$ for $e^{-\theta}$ in a $\text{Poisson}(\theta)$ model, by conditioning it on the sufficient statistic $T = \sum_{i=1}^n X_i$. Now, recall that $\text{Poisson}(\theta)$ is an exponential family, since the density can be written as

$$f_\theta(x) = \frac{1}{x!} \exp\{x \log \theta - \theta\}, \quad x = 0, 1, 2, \dots$$

Thus $T = \sum_i X_i$ is complete and sufficient. Therefore, the estimator from Lecture 9 that we found,

$$\mathbb{E}[U \mid T] = \left(1 - \frac{1}{n}\right)^T,$$

is the UMVUE of $e^{-\theta}$: it is an unbiased function of a complete sufficient statistic. We have now justified optimality among *all* unbiased estimators, not just improvement over the proportion of zeros.

Example 2 (Bernoulli sampling: correcting the plug-in estimator). Suppose $X_i \stackrel{\text{iid}}{\sim} \text{Ber}(\theta)$, $0 < \theta < 1$. We have

$$f_\theta(x) = \exp\left\{x \log \frac{\theta}{1-\theta} + \log(1-\theta)\right\}, \quad x \in \{0, 1\},$$

so $\text{Ber}(\theta)$ is an exponential family, and $T = \sum_i X_i$ is complete and sufficient. Since $\bar{X}_n = T/n$ is unbiased for θ , it is the UMVUE of θ .

Now suppose the target is $g(\theta) = \theta(1 - \theta)$. The plug-in estimator is $\bar{X}_n(1 - \bar{X}_n)$, but

$$\begin{aligned}\mathbb{E}_\theta[\bar{X}_n(1 - \bar{X}_n)] &= \mathbb{E}_\theta[\bar{X}_n] - \mathbb{E}_\theta[\bar{X}_n^2] \\ &= \theta - \left(\frac{\theta(1 - \theta)}{n} + \theta^2 \right) \\ &= \frac{n-1}{n} \theta(1 - \theta).\end{aligned}$$

Multiplying by $n/(n-1)$ gives an unbiased function of T . Hence the UMVUE is

$$\boxed{\frac{n}{n-1} \bar{X}_n(1 - \bar{X}_n) = \frac{T(n-T)}{n(n-1)}}.$$

No conditioning calculation was needed: once we found an unbiased function of the complete sufficient statistic, Lehmann–Scheffé did the rest.

Example 3 (Normal mean with known variance). Suppose $X_i \stackrel{\text{iid}}{\sim} \mathcal{N}(\theta, \sigma^2)$, where $\sigma^2 > 0$ is known. Write

$$f_\theta(x) = \frac{e^{-x^2/(2\sigma^2)}}{\sqrt{2\pi\sigma^2}} \exp\left\{ \frac{\theta}{\sigma^2}x - \frac{\theta^2}{2\sigma^2} \right\}.$$

Here $b(\theta) = \theta/\sigma^2$ and $R(x) = x$. So $\sum_i X_i$ is complete and sufficient. Since $\bar{X}_n = \frac{1}{n} \sum_i X_i$ is unbiased for θ , we conclude $\bar{X}_n$ is the UMVUE of θ . Let's now compute the UMVUE for θ^2 . Note that

$$\mathbb{E}_\theta[\bar{X}_n^2] = \theta^2 + \frac{\sigma^2}{n} \implies \boxed{\bar{X}_n^2 - \frac{\sigma^2}{n} \text{ is the UMVUE of } \theta^2}.$$

Example 4 (Poisson sampling: estimating a squared mean). Let $X_i \stackrel{\text{iid}}{\sim} \text{Poisson}(\theta)$ and $T = \sum_i X_i$. Since $T \sim \text{Poisson}(n\theta)$,

$$\mathbb{E}_\theta[T^2] = \text{Var}_\theta(T) + (\mathbb{E}_\theta[T])^2 = n\theta + n^2\theta^2.$$

Subtracting $\mathbb{E}_\theta[T] = n\theta$ gives

$$\mathbb{E}_\theta[T(T-1)] = n^2\theta^2 \implies \boxed{\frac{T(T-1)}{n^2} \text{ is the UMVUE of } \theta^2},$$

since it is an unbiased function of the complete sufficient statistic T .

11.2 Completeness and minimal sufficiency

Minimal sufficiency says we cannot compress a sufficient statistic further while keeping sufficiency. Completeness says there is at most one unbiased function of the statistic for each target. These are different claims, and as we show now, minimal sufficiency does not imply completeness.

Example 5 (Minimal sufficient, but not complete). Suppose $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Unif}(\theta, \theta+1)$, where $\theta \in \mathbb{R}$ and $n \geq 2$. In Homework 2, Problem 6(d), we proved that $T = (X_{(1)}, X_{(n)})$ is minimal sufficient. But T is not complete. To show this, we use that $\text{Unif}(\theta, \theta+1)$ is a location family. Write $X_i = \theta + Z_i$ with $Z_i \stackrel{\text{iid}}{\sim} \text{Unif}(0, 1)$. We know that $h(T) = X_{(n)} - X_{(1)} = Z_{(n)} - Z_{(1)}$ is *ancillary*: its distribution does not depend on θ . Thus for every θ , we have $\mathbb{E}_\theta[X_{(n)} - X_{(1)}] = c$ for some constant c (working out the math gives $c = (n-1)/(n+1)$). Therefore, the nonzero function $a(T) = X_{(n)} - X_{(1)} - c$ has $\mathbb{E}_\theta[a(T)] = 0$ for all θ , so T is not complete.

Proposition 1 (Using ancillarity to disprove completeness). If there exists a nonconstant ancillary statistic $A = A(T)$, then T is not complete.

Proof. Ancillarity makes $c = \mathbb{E}_\theta[A]$ independent of θ . Thus $a(T) = A(T) - c$ has mean zero for every θ , but is not almost surely zero. This contradicts completeness. $\square$

Example 6 (Normal sampling: the choice of statistic matters). Suppose $X_i \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2)$, with unknown variance $\sigma^2 > 0$. The statistic $S = \sum_i X_i$ is not complete: $a(S) = S$ has mean zero for every σ^2 , but $S \sim \mathcal{N}(0, n\sigma^2)$ is not almost surely zero.

On the other hand,

$$f_{\sigma^2}(x) = \frac{1}{\sqrt{2\pi}} \exp\left\{-\frac{x^2}{2\sigma^2} - \frac{1}{2} \log \sigma^2\right\}.$$

The natural statistic is x^2 . Hence $Q = \sum_i X_i^2$ is complete and sufficient. Since $\mathbb{E}_{\sigma^2}[Q] = n\sigma^2$,

$$\boxed{\frac{1}{n} \sum_{i=1}^n X_i^2 \text{ is the UMVUE of } \sigma^2.}$$

Theorem 2 (Complete and sufficient implies minimal sufficient). Suppose T is a statistic or a vector of statistics. If T is sufficient and complete, then it is minimal sufficient, using our convention that equalities may ignore events of probability zero.

Proof when T has finite variance. We give the proof assuming that T is univariate; the proof when T is a vector is similar. Let S be any sufficient statistic. We must show that T is a function of S . Define $g(\theta) = \mathbb{E}_\theta[T]$. Then T is an unbiased function of the complete sufficient statistic T , so Lehmann–Scheffé implies that T is the UMVUE of $g(\theta)$.

Since S is sufficient, Rao–Blackwell gives another unbiased estimator $\mathbb{E}[T | S]$ with

$$\text{Var}_\theta(\mathbb{E}[T | S]) \leq \text{Var}_\theta(T).$$

But T is the UMVUE, so equality holds, and total variance gives

$$0 = \text{Var}_\theta(T) - \text{Var}_\theta(\mathbb{E}[T | S]) = \mathbb{E}_\theta[\text{Var}_\theta(T | S)].$$

It follows that $\text{Var}_\theta(T | S) = \mathbb{E}_\theta[(T - \mathbb{E}[T | S])^2 | S] = 0$ with probability 1 for every θ , which implies $T = \mathbb{E}[T | S]$ with probability 1 for every θ . Sufficiency of S ensures that $\mathbb{E}[T | S]$ does not depend on θ . Thus T is a function of S , as required. $\square$

We therefore have

$$\boxed{\text{complete and sufficient} \implies \text{minimal sufficient},}$$

but the converse fails, as Example 5 shows. A complete sufficient statistic gives both the coarsest sufficient representation and uniqueness of unbiased functions.

Proposition 2 (Functions preserve completeness). If T is complete and $S = r(T)$, then S is complete.

Proof. If $\mathbb{E}_\theta[a(S)] = 0$ for every θ , then $\mathbb{E}_\theta[a(r(T))]$ is 0 for every θ . Apply completeness of T to the function $t \mapsto a(r(t))$ to obtain $a(S) = a(r(T)) = 0$ with probability 1 for every θ . $\square$

Coarsening preserves completeness, but it need not preserve sufficiency. For example, a constant statistic is complete, but generally not sufficient. A one-to-one transformation preserves both properties, since it retains exactly the same information.

11.3 A continuous conditioning example

Example 7 (Estimating an exponential survival probability). Let $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Exp}(\lambda)$. For a fixed $a > 0$, we want to estimate

$$g(\lambda) = \mathbb{P}_\lambda(X_1 > a) = e^{-\lambda a}.$$

Step 1: Find a complete sufficient statistic. We have that $\text{Exp}(\lambda)$ is an exponential family, since we may write the density as

$$f_\lambda(x) = \mathbb{1}_{\{x>0\}} \exp\{-\lambda x + \log \lambda\}.$$

So $T = \sum_i X_i$ is complete and sufficient.

Step 2: Find an unbiased starting estimator. Take $U = \mathbb{1}_{\{X_1 > a\}}$. Then $\mathbb{E}_\lambda[U] = e^{-\lambda a}$. We will compute $\mathbb{E}[U | T] = \mathbb{P}(X_1 > a | T)$.

Step 3: Find the conditional density. Assume $n \geq 2$. The variables X_1 and $V = \sum_{i=2}^n X_i$ are independent, with $V \sim \text{Gamma}(n-1, \lambda)$. Given $X_1 = x$, we have $T = x + V$. Independence means that conditioning on $X_1 = x$ does not change the distribution of V , so the conditional density of T is just the density of V shifted by x :

$$f_{T|X_1}(t | x) = f_V(t - x) = \frac{\lambda^{n-1}(t-x)^{n-2}e^{-\lambda(t-x)}}{\Gamma(n-1)}, \quad t > x.$$

Now use the joint-density identity and the definition of conditional density:

$$f_{X_1|T}(x | t) = \frac{f_{X_1,T}(x, t)}{f_T(t)} = \frac{f_{T|X_1}(t | x)f_{X_1}(x)}{f_T(t)}.$$

Since $T \sim \text{Gamma}(n, \lambda)$, for $0 < x < t$ this gives

$$\begin{aligned} f_{X_1|T}(x | t) &= \frac{\frac{\lambda^{n-1}(t-x)^{n-2}e^{-\lambda(t-x)}}{\Gamma(n-1)} \lambda e^{-\lambda x}}{\frac{\lambda^n t^{n-1} e^{-\lambda t}}{\Gamma(n)}} \\ &= \frac{n-1}{t} \left(1 - \frac{x}{t}\right)^{n-2}. \end{aligned}$$

The conditional density is zero otherwise. The unknown rate cancels, as sufficiency predicts.

Step 4: Condition and apply Lehmann–Scheffé. If $t \leq a$, then $X_1 < t \leq a$, so the conditional probability is zero. If $t > a$,

$$\mathbb{P}(X_1 > a | T = t) = \int_a^t \frac{n-1}{t} \left(1 - \frac{x}{t}\right)^{n-2} dx = \left(1 - \frac{a}{t}\right)^{n-1}.$$

Therefore,

$$\hat{g}(T) = \begin{cases} 0, & T \leq a, \\ \left(1 - \frac{a}{T}\right)^{n-1}, & T > a. \end{cases}$$

By the tower property, $\mathbb{E}_\lambda[\hat{g}(T)] = \mathbb{E}_\lambda[U] = g(\lambda)$. This estimator is an unbiased function of the complete sufficient statistic T , and is therefore the UMVUE.